

EXPT NO:08

MINI PROJECT

Financial Data Analytics

AIM:

To analyze Online Retail sales data using Python and SQL Server by managing a 20,000-record retail dataset, performing sales analysis using SQL queries, and creating visualizations to identify trends in sales, orders, countries, and product performance.

Software and Tools Required

- Python
- VS Code
- Pandas
- SQL Server
- SQL Server Management Studio (SSMS)
- Matplotlib

3. Process / Procedure

Step 1 — Project Setup

Create a project folder: **OnlineRetailAnalytics**

|

|— Online_Retail.xlsx

|— run.py

|— dashboard.py

|— Financial_20000.csv

|— Retail_Dashboard.png

The required Python libraries such as Pandas and Matplotlib are installed and used for data processing and visualization.

Step 2 — Create Database

Open SQL Server Management Studio (SSMS) and connect using:

Server: .\SQLEXPRESS

Authentication: Windows Authentication

Create the database:

```
CREATE DATABASE FinancialAnalytics;
```

```
GO
```

Step 3 — Prepare Dataset Using Python

The Online Retail dataset is loaded using Python. A subset of 20,000 records is selected and processed for analysis.

A new attribute named **TotalSales** is calculated using:

TotalSales = Quantity × UnitPrice

The processed dataset is saved as:

Financial_20000.csv

The dataset contains attributes such as:

- InvoiceNo
- StockCode

- Description
- Quantity
- InvoiceDate
- UnitPrice
- CustomerID
- Country
- TotalSales

Step 4 — Import Data into SQL Server

The processed CSV dataset is imported into SQL Server using SQL Server Management Studio.

The data flow is:

Online Retail Dataset



Python Data Processing



Financial_20000.csv



SQL Server



FinancialAnalytics



OnlineRetailData

The dataset containing 20,000 records is successfully stored in SQL Server.

Step 5 — Verify the Dataset

Run:

```
SELECT COUNT(*) AS TotalRecords
```

```
FROM OnlineRetailData;
```

Result:

TotalRecords

20000

This confirms that all 20,000 records have been successfully stored in SQL Server.

SQLQuery2.s...ous01 (64)*

```
1 SELECT
2     COUNT(*) AS TotalRecords,
3     COUNT(DISTINCT InvoiceNo) AS TotalTransactions,
4     ROUND(SUM(TotalSales), 2) AS TotalRevenue
5 FROM OnlineRetailData;
```

Results Messages

	TotalRecords	TotalTransactions	TotalRevenue
1	20000	1024	416073.23

Query executed successfully.

(local)\SQLEXPRESS (16.0 RTM) LAPTOP-BNIF9KDQ\anonym... FinancialAnalytics 00:00:00 Row: 1, Col: 1 1 rows

SQLQuery2.s...ous01 (64)*

```
1 SELECT TOP 10
2     Country,
3     ROUND(SUM(TotalSales), 2) AS TotalSales
4 FROM OnlineRetailData
5 GROUP BY Country
6 ORDER BY TotalSales DESC;
```

Results Messages

	Country	TotalSales
1	United Kingdom	370687.84
2	Germany	8056.94
3	Japan	7705.07
4	France	7382.32
5	IRE	5889.83
6	Norway	3787.12
7	Portugal	1976.47
8	Spain	1675.73
9	Lithuania	1661.06
10	Ovru	1590.82

Query executed successfully.

(local)\SQLEXPRESS (16.0 RTM) LAPTOP-BNIF9KDQ\anonym... FinancialAnalytics 00:00:00 Row: 1, Col: 1 10 rows

SQLQuery2.s...ous01 (64)*

```
SELECT TOP 10
    Description,
    SUM(Quantity) AS TotalQuantitySold
FROM OnlineRetailData
GROUP BY Description
ORDER BY TotalQuantitySold DESC;
```

Description	TotalQuantitySold
1 WORLD WAR 2 GLOVERB ASSORTD DESIGNS	4120
2 WHITE HANGING HEART T-LIGHT HOLDER	2445
3 MINI PAINT SET VINTAGE	2330
4 CREAM HEART CARD HOLDER	2213
5 BLACK HEART CARD HOLDER	2136
6 HAND WARMER EMBROIDSA DESIGN	2083
7 RED HARMONICA IN BOX	2029
8 ASSORTED COLOUR BIRD ORNAMENT	1915
9 ASSORTED COLOUR T-LIGHT HOLDER	1872
10 ROUND SNACK BOXES SET OF 4 FRUITS	1742

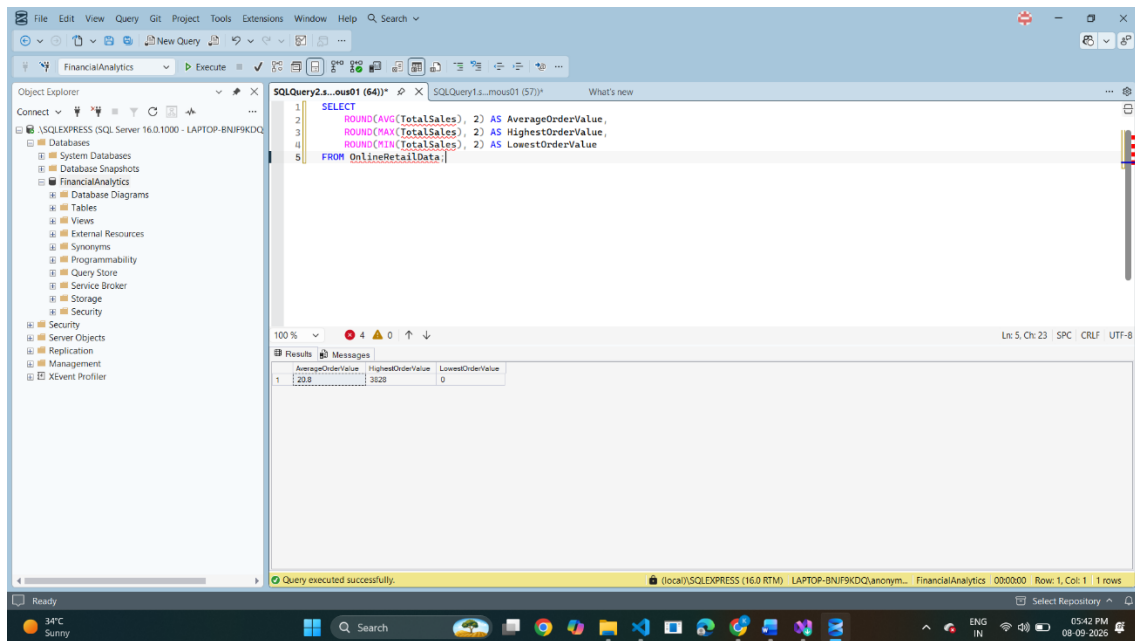
Query executed successfully.

SQLQuery2.s...ous01 (64)*

```
SELECT TOP 10
    Description,
    ROUND(SUM(TotalSales), 2) AS Revenue
FROM OnlineRetailData
GROUP BY Description
ORDER BY Revenue DESC;
```

Description	Revenue
1 REGENCY CAKESTAND 3 TIER	12974.4
2 VINTAGE UNION JACK MEMORBOARD	6769.54
3 WHITE HANGING HEART T-LIGHT HOLDER	6503.15
4 WOOD BLACK BOARD ANT WHITE FINISH	6154.38
5 CREAM HEART CARD HOLDER	5374.63
6 BLACK HEART CARD HOLDER	5090.64
7 ROUND SNACK BOXES SET OF 4 FRUITS	4476.5
8 BLACK RECORD COVER FRAME	4380.15
9 CHILLI LIGHTS	4105.44
10 RED WOOLLY HOTTIE WHITE HEART	3960.69

Query executed successfully.



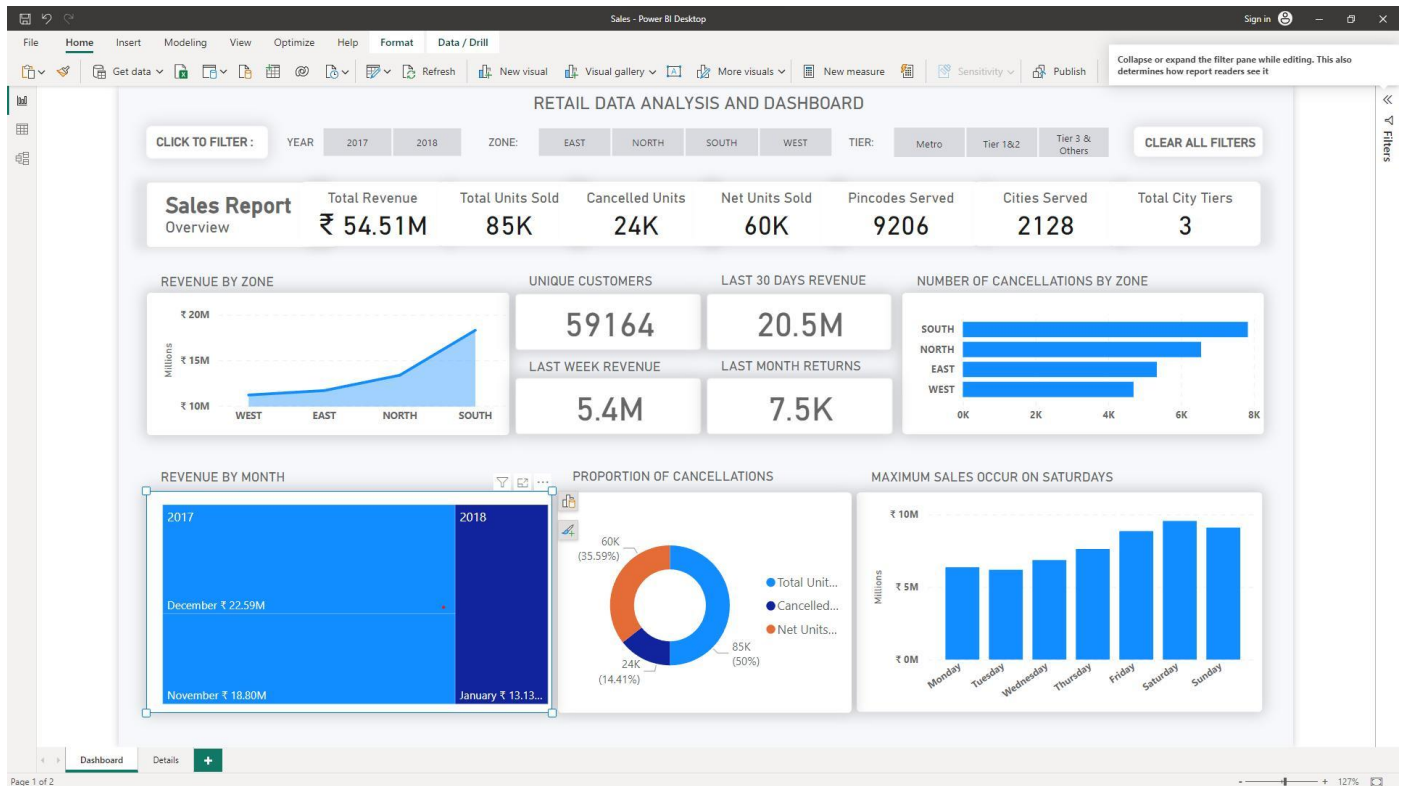
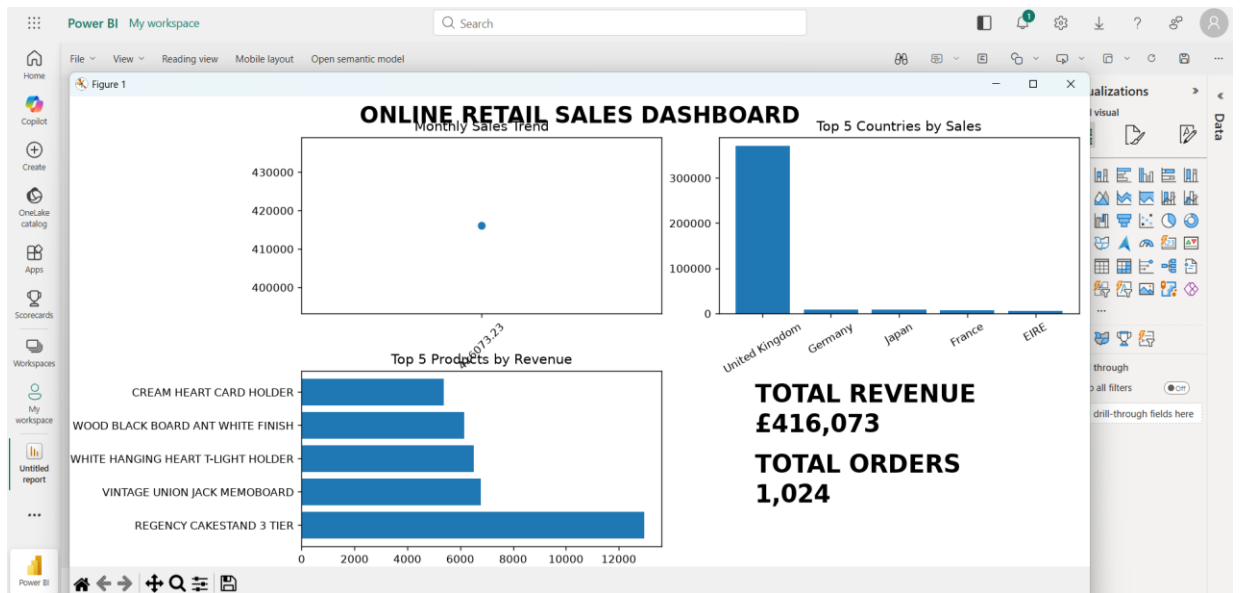
3. Data Visualization Using Python

After storing and verifying the dataset in SQL Server, Python and Matplotlib are used to create different visualizations.

The dataset is analyzed to generate:

- Monthly Sales Trend
- Top 5 Countries by Sales
- Top 5 Products by Revenue
- Total Revenue
- Total Orders

These visualizations are combined into an **Online Retail Sales Dashboard** to provide a clear overview of sales performance.



Python Code for Dashboard Generation:

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("Financial_20000.csv")

df["TotalSales"] = df["Quantity"] * df["UnitPrice"]
df["InvoiceDate"] = pd.to_datetime(df["InvoiceDate"])

df = df[df["Quantity"] > 0]

revenue = df["TotalSales"].sum()
orders = df["InvoiceNo"].nunique()

fig, ax = plt.subplots(2, 2, figsize=(14, 9))

fig.suptitle("ONLINE RETAIL SALES DASHBOARD",
             fontsize=20, fontweight="bold")

monthly = df.groupby(
    df["InvoiceDate"].dt.to_period("M")
)["TotalSales"].sum()

ax[0,0].plot(monthly.astype(str), monthly.values,
             marker="o")
ax[0,0].set_title("Monthly Sales Trend")

country = df.groupby("Country")["TotalSales"].sum().nlargest(5)

ax[0,1].bar(country.index, country.values)
ax[0,1].set_title("Top 5 Countries by Sales")

products = df.groupby(
    "Description"
)["TotalSales"].sum().nlargest(5)

ax[1,0].barh(products.index, products.values)
ax[1,0].set_title("Top 5 Products by Revenue")

ax[1,1].axis("off")

ax[1,1].text(
    0.1, 0.65,
    f"TOTAL REVENUE\n£{revenue:,.0f}",
    fontsize=22,
    fontweight="bold"
)

ax[1,1].text(
    0.1, 0.25,
    f"TOTAL ORDERS\n{orders:}",
    fontsize=22,
```

```
fontweight="bold"  
)  
  
plt.tight_layout()  
plt.savefig("Retail_Dashboard.png", dpi=200)  
plt.show()
```

OUTPUT:

The Python program successfully generated an Online Retail Sales Dashboard containing important business metrics and visualizations.

The dashboard displayed:

ONLINE RETAIL SALES DASHBOARD

Monthly Sales Trend

Top 5 Countries by Sales

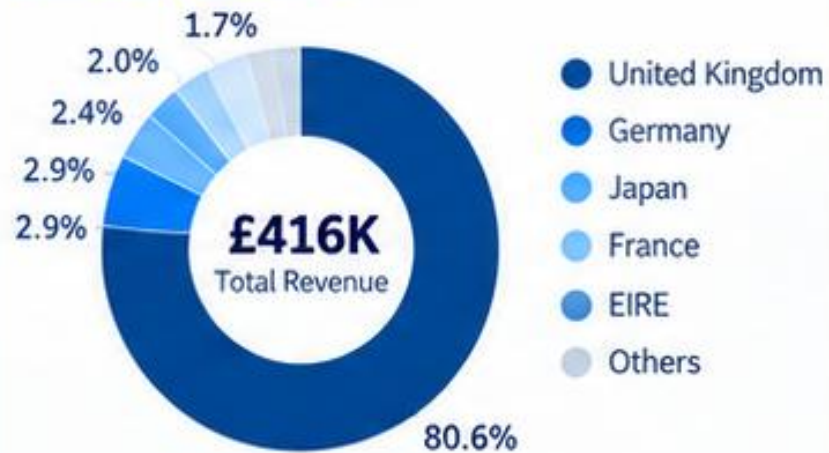
Top 5 Products by Revenue

Total Revenue: £416,073

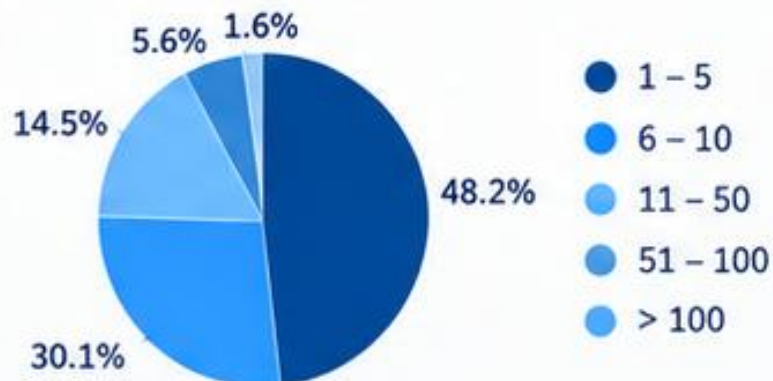
Total Orders: 1,024



Sales by Country



Order Quantity Distribution



Transactions by Day of Week

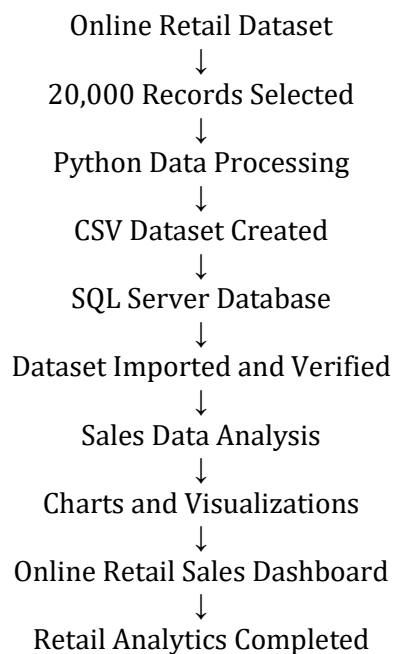


RESULT

The Online Retail Sales Analytics mini project was successfully implemented using Python, SQL Server, and Matplotlib. A dataset containing 20,000 retail transaction records was processed using Python and stored in CSV format. The dataset was imported into SQL Server and successfully queried for verification.

The retail data was further analyzed using Python to calculate sales revenue and identify important patterns. Visualizations were created to represent monthly sales trends, country-wise sales performance, top-performing products, total revenue, and total number of orders.

The final dashboard provides a consolidated view of the Online Retail dataset and helps in understanding overall sales performance.



CONCLUSION

Thus, the Online Retail dataset was successfully processed, stored, analyzed, and visualized using Python and SQL Server. The project demonstrates the application of data analytics and visualization techniques for understanding retail transaction data.

Overall, the mini project demonstrates how raw retail transaction data can be transformed into useful business insights through data processing, SQL analysis, and visualization techniques.